

JEE MAIN REVISION MASTER: ISOLATION OF ELEMENTS OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 106 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Isolation Of Elements — Applied Cases, Reactions & Traps	51 PYQs	Applied Topics
Very High	Isolation Of Elements — Standard Calculations & Problem Archetypes	43 PYQs	Quantitative Analysis
Very High	Isolation Of Elements — Core Principles & Governing Laws	15 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Isolation Of Elements — Applied Cases, Reactions & Traps	$600^{\circ}\text{C}, \text{C}, \text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2, \text{Al}_2\text{O}_3$
Isolation Of Elements — Standard Calculations & Problem Archetypes	$\text{Al}_2\text{O}_3, \text{Al}_2\text{O}_3, \Delta_r S^0$
Isolation Of Elements — Core Principles & Governing Laws	$\text{Na}_3\text{AlF}_6, \text{Al}_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Isolation Of Elements — Applied Cases, Reactions & Traps (Very High Yield — 51 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Isolation Of Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2025 Evening Shift] Q#1 (MCQ): "The purification method based on the following physical transformation is : $\underset{(X)}{\text{Solid}} \xrightarrow{\text{Heat}} \dots$ " [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Which one of the following reactions does not occur during extraction of copper?..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "Which one of the following is not an example of calcination?..." [Ans: B]

3.2 Isolation Of Elements — Standard Calculations & Problem Archetypes (Very High Yield — 43 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Isolation Of Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : In the Ellingham diagram,..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Which of the following is used as a stabilizer during the concentration of sulphide ores?..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A: In an Ellingham diagram, t..." [Ans: N/A]

3.3 Isolation Of Elements — Core Principles & Governing Laws (Very High Yield — 15 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Isolation Of Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Isolation Of Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Isolation Of Elements PYQs.

Representative Source PYQs:

- [2023 Morning Shift] Q#3 (MCQ): "Match List I with List II List-I List-II A. Reverberatory furnace I. Pig Iron B. Electrolytic cell II. Aluminium C. Blast furnace III. Silic..." [Ans: N/A]
- [2022 Evening Shift] Q#3 (MCQ): "The metal that is not extracted from its sulfide ore is :..." [Ans: N/A]
- [2022 Evening Shift] Q#3 (MCQ): "Match List I with List II. List-I Ore List-II Composition (A) Siderite (I) FeCO_3 (B) Malachite (II) $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$ (C) Sp..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 28th January Evening Shift (Q#1)

The purification method based on the following physical transformation is : $\text{Solid} \xrightarrow{\leq} \text{Vapour} \xrightarrow{\leq} \text{Solid}$

- (A) Extraction
- (B) Crystallization
- (C) Distillation
- (D) Sublimation

Correct Answer: None

Solution Analysis: The purification method described by the transformation: $\text{Solid} \xrightarrow{\leq} \text{Vapour} \xrightarrow{\leq} \text{Solid}$ is known as Sublimation . In sublimation, a solid substance directly transforms into its gaseous phase upon heating and reverts back to the solid phase upon cooling. This technique is commonly used to purify substances, particularly those that sublime more easily than others under the same conditions. Sublimation is effective for separating substances with diff

Benchmark Problem 2: JEE Main 2023 (Online) 12th April Morning Shift (Q#1)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : In the Ellingham diagram, a sharp change in slope of the line is observed for $\text{Mg} \rightarrow \text{MgO}$ at $\sim 1120^\circ\text{C}$ Reason R: There is a large change of entropy associated with the change of state In the light of the above statements, choose the correct answer from the options given below

- (A) Both A and R are true and R is the correct explanation of A
- (B) A is false but R is true
- (C) Both A and R are true but R is NOT the correct explanation of A
- (D) A is true but R is false

Correct Answer: None

Solution Analysis: In the Ellingham diagram, a sharp change in slope of the line is observed for Mg-MgO at $\sim 1120^\circ\text{C}$ because that is the boiling point of magnesium. There is a large increase in entropy associated with the change of state of magnesium. So, both Assertion (A) and Reason (R) are true and (R) is the correct explanation of (A) .

Benchmark Problem 3: JEE Main 2023 (Online) 24th January Morning Shift (Q#3)

Match List I with List II List-I List-II A. Reverberatory furnace I. Pig Iron B. Electrolytic cell II. Aluminium C. Blast furnace III. Silicon D. Zone Refining furnace IV. Copper Choose the correct answer from the options given below :

- (A) -III, B-IV, C-I, D-II
- (B) -IV, B-II, C-I, D-III
- (C) -I, B-IV, C-II, D-III
- (D) -I, B-III, C-II, D-IV

Correct Answer: None

Solution Analysis: (A) Reverberatory furnace is used for extraction of copper. (B) Electrolytic cell is used for obtaining highly reactive metals like aluminium. (C) Blast furnace is used for pig iron. (D) Zone refining is used for purification of silicon.